LOCAL-PROMPT: EXTENSIBLE LOCAL PROMPTS FOR FEW-SHOT OUT-OF-DISTRIBUTION DETECTION

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ABSTRACT

Few-shot out-of-distribution (OOD) detection with vision-language models (VLMs) is challenging because hard OOD samples often differ from in-distribution (ID) classes only in subtle, localized regions. Existing prompt learning methods optimize global text embeddings and neglect fine-grained regional cues, limiting their robustness under scarce supervision. In this work, we propose Local-Prompt, a coarse-to-fine few-shot OOD detection framework that decomposes prompt learning into frozen global prompts and trainable local prompts. Local-Prompt operates in two stages: (1) *Global Prompt Guided Negative Augmentation*, which applies random crops to training images and uses frozen global prompts to select the hardest positives and negatives, injecting local outlier knowledge without requiring real OOD data; and (2) *Local Prompt Enhanced Regional Regularization*, which trains class-specific local prompts and a bank of local negative prompts via a contrastive loss over top-k local regions and a diversity regularization loss on negative prompts. At inference, we introduce a Regional-MCM score that aggregates global MCM with region-level scores, incorporating both local positive and negative prompts. Our framework is orthogonally extensible: trained global prompts from existing methods can be seamlessly plugged in to further boost detection performance. On the ImageNet-1k benchmark (4-shot), Local-Prompt+LoCoOp achieves an average FPR95 of 26.29 and AUROC of 94.03, surpassing prior 16-shot methods. On near-OOD benchmarks and ImageNet-100, Local-Prompt consistently outperforms state-of-the-art zero-shot, fully supervised, and few-shot competitors.

1 INTRODUCTION

Deploying machine learning systems in the open world requires reliable detection of out-of-distribution (OOD) samples—inputs that come from classes unseen during training (Hendrycks & Gimpel, 2016; Liu et al., 2020). The rise of large-scale vision-language models (VLMs) such as CLIP (Radford et al., 2021) has opened new avenues for OOD detection, where rich multi-modal representations enable zero-shot detection without any labeled data (Ming et al., 2022; Miyai et al., 2023b). However, as deployment scenarios become more demanding, there is increasing interest in the *few-shot* regime: using only a handful of labeled ID examples per class to significantly improve over zero-shot baselines.

Recent prompt learning methods for few-shot OOD detection, such as CoOp (Zhou et al., 2022b) and LoCoOp (Miyai et al., 2023a), learn *global* text prompts that embed ID class semantics. While effective for coarse-level separation between ID and OOD distributions, these global approaches overlook a critical failure mode: *hard OOD samples that differ from ID classes only in subtle, localized image regions*. As illustrated in Figure 1, an image may be globally similar to an ID class while containing a small but decisive local region that betrays its OOD nature—a nuance that global features fundamentally cannot capture.

This observation motivates our key insight: *local regions carry discriminative signals that can separate subtle OOD cases when global features are insufficient*. By focusing prompt learning exclusively on local regions with hard negative augmentation, OOD detection becomes both more precise and more sample-efficient.

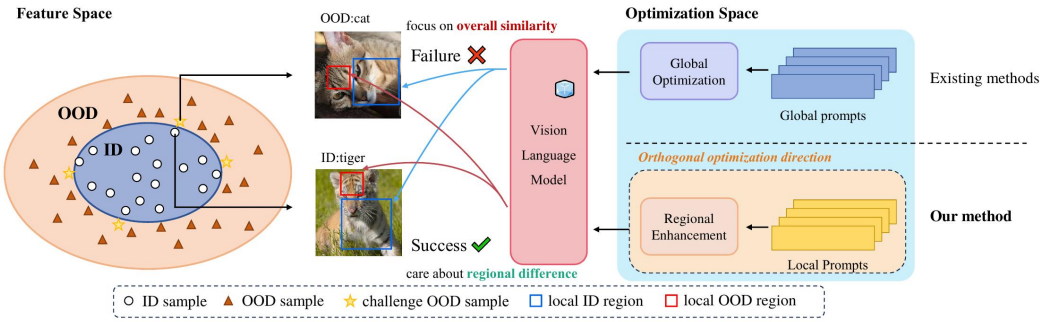

Figure 1: Comparison of prompt learning for OOD detection task. Global prompt methods embed entire-image features and may fail on hard OOD samples where only localized regions differ. Local-Prompt trains local prompts to capture regional discriminative signals for robust detection.

We propose **Local-Prompt**, a coarse-to-fine few-shot OOD detection framework that adopts a principled decomposition of prompt learning into global (frozen) and local (trainable) components. The design philosophy of Local-Prompt rests on two pillars:

(1) Global Prompt Guided Negative Augmentation. Rather than training global prompts from scratch with local regions (which risks interfering with coarse-level ID semantics already captured by the pre-trained CLIP representation), we fix the global prompts and use them as an oracle to identify *hard positives* and *hard negatives* among random crops of training images. Hard positives are local regions most similar to their GT class embedding; hard negatives are local regions that look globally dissimilar. This provides a natural and inexpensive source of local OOD knowledge without requiring real OOD samples.

(2) Local Prompt Enhanced Regional Regularization. We train class-specific local prompts alongside a shared bank of local negative prompts using a contrastive loss applied over the top-k most discriminative local regions. A diversity loss encourages negative prompts to cover diverse OOD semantics. At inference, our *Regional-MCM* score combines the global MCM score with a regional score computed over local features, incorporating both local positive and negative prompts for a balanced ID/OOD assessment.

A key strength of Local-Prompt is its *extensibility*: since global prompts are kept frozen during training, any separately trained global prompt (e.g., from LoCoOp) can be plugged in at inference to provide complementary coarse-level signals. This orthogonal combination consistently yields further gains.

Our main contributions are:

- We introduce a coarse-to-fine few-shot OOD detection paradigm that freezes global prompts and trains local prompts to capture regional outlier cues, enabling more precise detection of hard OOD samples.
- We propose *Global Prompt Guided Negative Augmentation* to construct hard positive/negative local pairs from training images without external OOD data, and *Local Prompt Enhanced Regional Regularization* with contrastive top-k regional loss and diversity loss on local negative prompts.
- We develop *Regional-MCM*, a novel inference-time OOD score that integrates global and regional evidence through local positive and negative prompts.
- We demonstrate orthogonal extensibility: combining Local-Prompt with LoCoOp global prompts further improves performance, establishing state-of-the-art results on large-scale OOD detection benchmarks.

Extensive experiments on ImageNet-1k, ImageNet-100, and near-OOD benchmarks validate the effectiveness of Local-Prompt, which surpasses prior 16-shot methods in the 4-shot regime and achieves superior near-OOD detection performance against all baselines.

2 RELATED WORK

OOD Detection with Pre-trained Models. Traditional OOD detection methods rely on scores derived from a softmax classifier trained on ID data (Hendrycks & Gimpel, 2016; Liu et al., 2020; Sun et al., 2022). With the advent of powerful pre-trained VLMs, zero-shot OOD detection became feasible: MCM (Ming et al., 2022) aligns image features with textual class descriptions to produce an OOD score directly from CLIP representations. GL-MCM (Miyai et al., 2023b) extends this by combining global and local CLIP features. NegLabel (Jiang et al., 2024) further improves zero-shot OOD detection by incorporating a large set of negative labels from external corpora, achieving strong performance at the cost of requiring outlier samples.

Prompt Learning for Few-Shot Adaptation. CoOp (Zhou et al., 2022b) pioneered learnable context vectors for adapting CLIP to downstream classification tasks with few labeled examples. CoCoOp (Zhou et al., 2022a) extends CoOp with instance-conditioned prompts for better generalization. ID-like (Bai et al., 2024) discovers ID-like OOD samples in the vicinity of ID classes to improve few-shot OOD detection. CLIPN (Wang et al., 2023) incorporates negation-semantic prompts. The most closely related work is LoCoOp (Miyai et al., 2023a), which performs local regularization during prompt training by pushing local CLIP features away from ID class embeddings. However, LoCoOp trains *global* prompts with local regularization, whereas Local-Prompt freezes global prompts and trains *dedicated* local prompts with hard negative augmentation and regional contrastive learning.

Local Features for OOD Detection. Several works have explored local or regional features for OOD detection. Zhang et al. (Zhang et al., 2023) propose a multi-scale detection framework combining global and local representations through attention-based local propagation. LoCoOp (Miyai et al., 2023a) implicitly leverages local CLIP features by treating background crops as OOD signals during training. Local-Prompt explicitly trains local prompts against top-k regional features guided by hard negatives, providing a more principled and data-efficient local learning mechanism.

Outlier Exposure and Negative Sampling. Outlier Exposure (Hendrycks et al., 2018) uses auxiliary OOD data during training to improve detection. NPOS (Tao et al., 2022) synthesizes OOD features non-parametrically in feature space. Textual-OE (Park et al., 2023) explores textual outliers. In contrast, Local-Prompt generates local pseudo-OOD samples by augmenting training images with random crops, requiring no external OOD data.

3 PRELIMINARY

CLIP and Visual Prompts. CLIP (Radford et al., 2021) is a vision-language model trained on 400M image-text pairs via contrastive learning. Given an image x and a set of class descriptions $\{t_c\}_{c=1}^{C}$, CLIP computes cosine similarities between the image embedding $f(x)$ and text embeddings $\{g(t_c)\}$. For OOD detection, the Maximum Concept Matching (MCM) score (Ming et al., 2022) is defined as:

$$S_{\mathrm{MCM}}(x) = -\max_{c} \frac{\exp(\cos(f(x), g(t_c))/T)}{\sum_{c'} \exp(\cos(f(x), g(t_{c'}))/T)}, \tag{1}$$

where T is a temperature parameter. Samples with high S_{MCM} are flagged as OOD.

CLIP's Vision Transformer (ViT) backbone additionally provides local (patch-level) features $\{f_j(x)\}_{j=1}^{N}$ from intermediate attention outputs, where N is the number of patches. GL-MCM (Miyai et al., 2023b) combines global and local CLIP scores for improved zero-shot OOD detection.

Prompt Learning. Context Optimization (CoOp) (Zhou et al., 2022b) replaces hand-crafted text prompts with learnable context vectors $\mathbf{v} = [v_1, \ldots, v_M]$, giving prompts of the form $t_c = [v_1, \ldots, v_M, \text{classname}_c]$. During few-shot training, only $\mathbf{v}$ is optimized while CLIP's image and text encoders are frozen, using a cross-entropy loss on ID training images. LoCoOp (Miyai et al., 2023a) augments CoOp's training with a local OOD regularization term that uses local CLIP features of background crops as implicit negative samples.

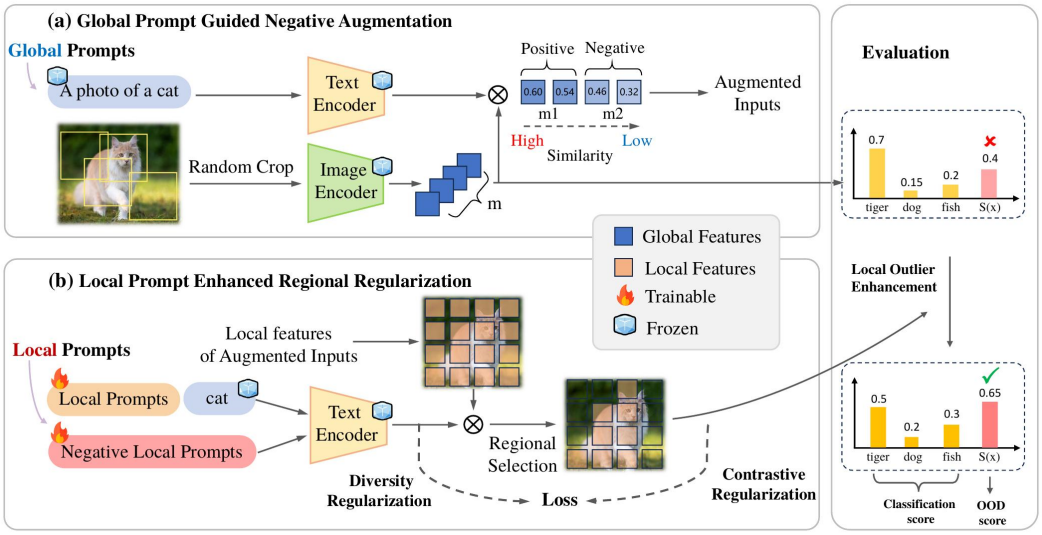

Figure 2: Detailed structure of the proposed Local-Prompt. Left: Global Prompt Guided Negative Augmentation selects hard positive/negative crops using frozen global prompts. Right: Local Prompt Enhanced Regional Regularization trains local prompts with contrastive loss over top-k regions.

Few-Shot OOD Detection Setting. In few-shot OOD detection, we have access to K labeled images per class from the ID dataset $\mathcal{D}_{\text{ID}} = \{(x_i, y_i)\}$ with C classes, but no labeled OOD data. The goal is to learn a detection score $S(x)$ such that ID samples receive low scores and OOD samples receive high scores, maximizing AUROC and minimizing FPR95 (false positive rate at 95% true positive rate).

4 METHODOLOGY

4.1 OVERVIEW

Local-Prompt decomposes prompt learning into two complementary components:

1. **Frozen global prompts** $\{\mathbf{p}_c^g\}_{c=1}^C$: initialized as "a photo of a {classname}" and kept frozen throughout training. These capture coarse-level ID class semantics via CLIP's pre-trained alignment.
2. **Trainable local prompts** $\{\mathbf{p}_c^l\}_{c=1}^C$ and **local negative prompts** $\{\mathbf{p}_k^n\}_{k=1}^{N_{\text{neg}}}$: class-specific local prompts encode fine-grained regional ID features, while the shared bank of local negative prompts captures diverse OOD semantics.

The training procedure proceeds in two stages per mini-batch, as illustrated in Figure 2. The overall loss is:

$$\mathcal{L} = \mathcal{L}_{\text{pos}} + \lambda \mathcal{L}_{\text{neg}} + \mu \mathcal{L}_{\text{reg}}, \tag{2}$$

where $\mathcal{L}_{\text{pos}}$ is the positive contrastive loss, $\mathcal{L}_{\text{neg}}$ is the negative prompt contrastive loss, $\mathcal{L}_{\text{reg}}$ is the diversity regularization, and λ, μ are balancing weights.

4.2 GLOBAL PROMPT GUIDED NEGATIVE AUGMENTATION

A key challenge in few-shot OOD detection is the absence of labeled OOD data. To obtain informative local OOD signals from training images, we propose *Global Prompt Guided Negative Augmentation*.

Given a training image x with label y, we generate multiple augmented views $\{x^{(m)}\}_{m=1}^M$ by applying random crops of varying scales and aspect ratios. Each crop $x^{(m)}$ focuses on a different spatial

region and may contain varying amounts of background (OOD-like content) and foreground (ID content).

We use the frozen global text embeddings $\mathbf{g}_y = g(\mathbf{p}_y^g)$ to assess the ID-relevance of each crop:

$$s^{(m)} = \cos\big(f(x^{(m)}),\, \mathbf{g}_y\big), \tag{3}$$

where $f(\cdot)$ is CLIP's image encoder. *Hard positives* are the crops with the highest similarity scores (most ID-like local regions), and *hard negatives* are the crops with the lowest similarity (most OOD-like local regions). Formally, we select:

$$x^+ = \arg\max_m s^{(m)}, \quad x^- = \arg\min_m s^{(m)}. \tag{4}$$

This selection strategy is principled: using frozen global prompts as a semantic filter ensures that the hard positives/negatives are defined consistently with the global ID semantics, while avoiding the instability that would arise from using the still-training local prompts. Crucially, this approach requires no external OOD dataset—hard negatives are derived entirely from the training images themselves.

4.3 Local Prompt Enhanced Regional Regularization

Given the hard positive image x^+ and hard negative image x^- selected by the global prompt guided augmentation, we extract their local (patch-level) features:

$$\mathbf{F}^+ = \{f_j(x^+)\}_{j=1}^N, \quad \mathbf{F}^- = \{f_j(x^-)\}_{j=1}^N, \tag{5}$$

where N is the number of ViT patch tokens.

Class-specific Local Prompts. The local prompts $\{\mathbf{p}_c^l\}$ are class-specific learnable context vectors forming prompts $t_c^l = [\mathbf{p}_c^l, \text{classname}_c]$. Their text embeddings $\mathbf{w}_c^l = g(t_c^l)$ are trained to align with the most discriminative local regions of ID images.

Positive Contrastive Loss $\mathcal{L}_{\text{pos}}$. For each training example (x^+, y), we compute the similarity between each local feature $f_j(x^+)$ and the local text embedding $\mathbf{w}_y^l$ of the ground-truth class. We identify the top-k most informative local regions via:

$$\mathcal{R}^+(y) = \text{Top-}k\big(\{\cos(f_j(x^+), \mathbf{w}_y^l)\}_{j=1}^N\big). \tag{6}$$

The positive contrastive loss encourages these regions to be most similar to the correct local prompt:

$$\mathcal{L}_{\text{pos}} = -\frac{1}{B}\sum_{b=1}^{B} \log \frac{\sum_{j\in\mathcal{R}^+} \exp(f_j^+ \cdot \mathbf{w}_y^l / T)}{\sum_{j\in\mathcal{R}^+}\sum_{c=1}^{C} \exp(f_j^+ \cdot \mathbf{w}_c^l / T)}. \tag{7}$$

Negative Contrastive Loss $\mathcal{L}_{\text{neg}}$. The local negative prompts $\{\mathbf{p}_k^n\}_{k=1}^{N_{\text{neg}}}$ are designed to capture OOD semantics. Using the hard negative images x^-, we train these prompts to be most similar to OOD-like local regions. For the top-k OOD-like local features from x^-, we maximize agreement with the negative prompt bank:

$$\mathcal{L}_{\text{neg}} = -\frac{1}{B}\sum_{b=1}^{B} \log \frac{\sum_{j\in\mathcal{R}^-}\sum_{k=1}^{N_{\text{neg}}} \exp(f_j^- \cdot \mathbf{w}_k^n / T)}{\sum_{j\in\mathcal{R}^-}\sum_{c=1}^{C+N_{\text{neg}}} \exp(f_j^- \cdot \mathbf{w}_c / T)}, \tag{8}$$

where $\mathcal{R}^-$ selects the top-k most negative-like local features from x^-, and $\mathbf{w}_c$ ranges over both local class prompts and negative prompts.

Diversity Regularization $\mathcal{L}_{\text{reg}}$. To prevent the negative prompts from collapsing into a single mode, we impose a diversity regularization:

$$\mathcal{L}_{\text{reg}} = \frac{1}{N_{\text{neg}}^2}\sum_{k=1}^{N_{\text{neg}}}\sum_{k'=1}^{N_{\text{neg}}} \cos(\mathbf{w}_k^n, \mathbf{w}_{k'}^n). \tag{9}$$

This penalizes high mutual similarity among negative prompts, encouraging each to specialize in a different aspect of OOD semantics.

4.4 REGIONAL-MCM SCORE

At inference, Local-Prompt computes a composite OOD score that integrates global and local evidence. Given a test image x:

Global MCM score:

$$S_{\text{global}}(x) = -\max_{c} \frac{\exp(\cos(f(x), \mathbf{g}_c)/T)}{\sum_{c'} \exp(\cos(f(x), \mathbf{g}_{c'})/T)}. \tag{10}$$

Regional local score: We compute local features $\{f_j(x)\}$ and evaluate them against both local class prompts $\{\mathbf{w}_c^l\}$ and local negative prompts $\{\mathbf{w}_k^n\}$. The regional score uses a softmax over the combined prompt set:

$$p_j^c(x) = \frac{\exp(f_j \cdot \mathbf{w}_c^l/T)}{\sum_{c'} \exp(f_j \cdot \mathbf{w}_{c'}^l/T) + \sum_k \exp(f_j \cdot \mathbf{w}_k^n/T)}. \tag{11}$$

The top-k regions are aggregated to form the regional MCM score:

$$S_{\text{local}}(x) = -\text{Mean}\big(\text{Top-}k\big(\{\max_{c} p_j^c(x)\}_{j=1}^N\big)\big). \tag{12}$$

The final **Regional-MCM** score combines both signals:

$$S_{\text{R-MCM}}(x) = S_{\text{global}}(x) + S_{\text{local}}(x). \tag{13}$$

Extensibility with Global Prompts. When a trained global prompt (e.g., from LoCoOp) is available, we substitute the frozen global prompts with the trained ones in S_{global}, while the local prompts remain as trained by Local-Prompt. This plug-and-play combination consistently provides additional gains at no extra training cost.

The ID classification score uses a joint global-local prediction:

$$S_{\text{ID}}(x) = \exp(f(x) \cdot \mathbf{g}_y/T) \cdot \text{Mean}\big(\text{Top-}k\big(\{\exp(f_j \cdot \mathbf{w}_y^l/T)\}_{j=1}^N\big)\big), \tag{14}$$

which improves both OOD detection and ID classification accuracy.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Datasets. We evaluate on two standard large-scale benchmarks:

- **ImageNet-1k** (Dosovitskiy et al., 2020): 1,000 ID classes. OOD test sets: iNaturalist (Van Horn et al., 2018), SUN (Xiao et al., 2010), Places (Zhou et al., 2017), and Texture (Cimpoi et al., 2014).
- **ImageNet-100**: A 100-class subset of ImageNet-1k as ID, with the same four OOD test sets.
- **Near-OOD**: ImageNet-10 and ImageNet-20 are used as paired ID/OOD datasets to evaluate detection of semantically close OOD classes.

Evaluation Metrics. We report FPR95 (false positive rate at 95% true positive rate; lower is better) and AUROC (higher is better), as well as ID classification accuracy.

Baselines. We compare against zero-shot methods (MCM (Ming et al., 2022), GL-MCM (Miyai et al., 2023b)), fully-supervised methods (MSP (Hendrycks & Gimpel, 2016), Energy (Liu et al., 2020), KNN (Sun et al., 2022), NPOS (Tao et al., 2022), Textual-OE (Park et al., 2023)), few-shot methods (CoOp (Zhou et al., 2022b), LoCoOp (Miyai et al., 2023a)), and NegLabel (Jiang et al., 2024) which leverages powerful external outlier samples.

Table 1: Results when ImageNet-1k is used as ID data. We compare the methods in different tuning manners. † represents the result by our re-implementation. **Bold** values represent the best results.

Method	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
Zero-shot										
MCM	30.91	94.61	37.59	92.57	44.69	89.77	57.77	86.11	42.74	90.77
GL-MCM	15.18	96.71	30.42	93.09	38.85	89.90	57.93	83.63	35.47	90.83
Full-tuning										
MSP	54.05	87.43	73.37	78.03	72.98	78.03	68.85	79.06	67.31	80.64
Energy	29.75	94.68	53.18	87.33	56.40	85.60	51.35	88.00	47.67	88.90
KNN	29.17	94.52	35.62	92.67	39.61	91.02	64.35	85.67	42.19	90.97
NPOS	16.58	96.19	43.77	90.44	45.27	89.44	46.12	88.80	37.93	91.22
Textual-OE	29.61	94.74	57.12	87.34	66.82	83.71	79.29	77.76	58.21	85.88
Few-shot					4-*shot*					
CoOp	18.95	95.52	29.58	92.90	38.72	89.64	48.03	85.87	33.82	90.98
Local-Prompt	**9.65**	**97.87**	20.40	95.57	29.39	92.67	51.20	88.00	27.66	93.53
LoCoOp†	21.67	95.69	22.98	95.07	31.41	92.10	49.79	87.85	31.46	92.68
Local-Prompt+LoCoOp	12.81	97.29	**19.34**	**95.85**	**27.53**	**92.97**	**45.51**	**89.99**	**26.29**	**94.03**
					16-*shot*					
CoOp	14.60	96.62	28.48	92.65	36.49	89.98	43.13	88.03	30.67	91.82
Local-Prompt	8.71	98.10	23.97	94.85	32.50	92.32	47.93	89.04	28.27	93.58
LoCoOp	16.05	96.86	23.44	95.07	32.87	91.98	42.28	90.19	28.66	93.52
Local-Prompt+LoCoOp	**8.63**	**98.07**	**23.23**	**95.12**	**31.74**	**92.42**	**34.50**	**92.29**	**24.52**	**94.48**

Table 2: ID accuracy evaluation using ImageNet-1k as ID data.

Method	Tuning Manner	ID accuracy
MCM	*zero-shot*	67.01
GL-MCM		67.01
KNN	*full-tuning*	79.64
NPOS		79.42
LoCoOp	4-*shot*	69.50
Local-Prompt		**70.52**
LoCoOp	16-*shot*	71.86
Local-Prompt		**74.24**

Table 3: Comparison results of near OOD detection tasks. We report 16-shot results for few-shot methods. Our method can achieve performance against other methods.

Method	ID OOD	ImageNet-10 ImageNet-20		ImageNet-20 ImageNet-10		**Average**	
		FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MCM		5.00	98.71	12.51	97.70	8.75	98.21
GL-MCM		10.10	98.04	9.00	98.62	9.55	98.33
LoCoOp		11.20	97.49	12.00	97.79	11.60	97.64
Local-Prompt		**3.90**	**99.06**	**6.20**	**98.84**	**5.05**	**98.95**

Implementation Details. We use CLIP ViT-B/16 as our backbone. For few-shot experiments, we evaluate 4-shot and 16-shot settings. The number of local negative prompts N_{neg} is set to 200 (ablated in Section 5.6). The top-k parameter for regional aggregation is set to $k = 10$. The temperature T is 0.01. Loss balancing weights $\lambda = 1.0$ and $\mu = 0.1$. We use random crop augmentation with scales in $[0.08, 1.0]$ for negative augmentation.

5.2 Main Results on ImageNet-1k

Table 1 presents comprehensive results on the ImageNet-1k benchmark. Several observations stand out:

4-shot setting. Local-Prompt alone achieves an average FPR95 of 27.66 and AUROC of 93.53, outperforming CoOp (FPR95 33.82, AUROC 90.98) by a large margin. Local-Prompt particularly excels on iNaturalist (9.65 FPR95 vs. 18.95 for CoOp), a dataset with many subtle biological specimens that exhibit local-level differences. Combined with LoCoOp's global prompts, Local-Prompt+LoCoOp achieves average FPR95 of 26.29 and AUROC of 94.03, surpassing even 16-shot LoCoOp (FPR95 28.66, AUROC 93.52) with only 4 labeled examples per class.

16-shot setting. The trend continues at 16 shots, where Local-Prompt+LoCoOp reaches FPR95 24.52 and AUROC 94.48, representing further improvement. Notably, Local-Prompt alone (FPR95 28.27) already outperforms LoCoOp (FPR95 28.66) at the same shot count.

Comparison with fully-supervised methods. Despite using only 4–16 labeled examples, Local-Prompt variants outperform most fully-supervised methods. For example, NPOS (37.93 FPR95) and KNN (42.19 FPR95) use the full ImageNet training set yet fall behind our 4-shot results.

ID Classification Accuracy. Table 2 shows that Local-Prompt achieves superior ID accuracy alongside improved OOD detection. In the 4-shot setting, Local-Prompt attains 70.52% accuracy vs. 69.50% for LoCoOp (+1.02%). In the 16-shot setting, the gap widens to +2.38% (74.24% vs. 71.86%). The joint global-local scoring function is responsible for this gain: by incorporating regional features in classification, Local-Prompt makes more precise ID predictions.

Table 4: Experiments on ImageNet-100. 4-shot results are reported for few-shot tuning.

Method	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MCM	18.13	96.77	36.45	94.54	34.52	94.36	41.22	92.25	32.58	94.48
LoCoOp†	44.94	93.31	30.70	94.70	34.31	93.93	55.92	90.01	41.47	92.99
Local-Prompt	**6.99**	**98.05**	**24.34**	**96.03**	**26.04**	**95.49**	**36.32**	**93.67**	**23.42**	**95.81**

Near-OOD Detection. The right half of Table 2 reports near-OOD results on ImageNet-10/20 (16-shot). Local-Prompt achieves an average FPR95 of 5.05 and AUROC of 98.95, substantially outperforming GL-MCM (9.55/98.33) and LoCoOp (11.60/97.64). Near-OOD detection is especially challenging because OOD classes are semantically close to ID classes. The fact that Local-Prompt excels here confirms that local prompt learning captures the subtle discriminative features that distinguish near-OOD samples.

5.3 Results on ImageNet-100

Table 4 shows results on the ImageNet-100 benchmark (4-shot). Local-Prompt achieves average FPR95 of 23.42 and AUROC of 95.81, substantially outperforming MCM (32.58/94.48) and LoCoOp (41.47/92.99). The large improvement over LoCoOp on ImageNet-100 suggests that local prompt learning is especially advantageous when the ID class space is constrained, where distinguishing subtle within-domain differences becomes critical.

5.4 Comparison with NegLabel

Table 5 compares Local-Prompt against NegLabel (Jiang et al., 2024), which uses a powerful external database of negative labels as outlier samples—information not available to our method. Despite this significant advantage for NegLabel, Local-Prompt achieves competitive or superior average performance: Local-Prompt (FPR95 24.52, AUROC 94.48) vs. NegLabel (FPR95 25.40, AUROC 94.21) on the ImageNet-1k benchmark. This demonstrates that our local prompt learning can match the effectiveness of strong outlier exposure methods while requiring no external OOD data.

5.5 Visualization

Figure 3 visualizes the regional attention density maps comparing Local-Prompt against global prompt methods on iNaturalist OOD samples. The maps confirm that Local-Prompt's local prompts focus attention on the most discriminative image regions (e.g., subtle texture differences in feathers or skin patterns), whereas global methods spread attention uniformly.

Figure 4 visualizes ID and OOD related regions as identified by the trained local prompts. The local positive prompts (green) highlight foreground object features, while the local negative prompts (red) highlight background and atypical regions. Figure 5 shows qualitative examples where global-based methods fail: scene images from SUN with subtle structural differences that are imperceptible at the global level but clearly visible in localized regions. Local-Prompt successfully detects these cases by capturing regional cues.

5.6 Ablation Studies

Ablation on Loss Components. Table 6 ablates the contribution of each loss component. Starting from the baseline with only $\mathcal{L}_{\text{pos}}$ (FPR95 30.85), adding $\mathcal{L}_{\text{neg}}$ reduces FPR95 to 29.27 (-1.58), and further adding $\mathcal{L}_{\text{reg}}$ achieves FPR95 28.77 (-0.50). All three components contribute positively, confirming the complementary nature of positive contrastive learning, negative prompt training, and diversity regularization.

Table 5: Comparison with NegLabel, which leverages powerful outlier samples.

Method	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
NegLabel (w outlier)	1.91	99.49	20.53	95.49	35.59	91.64	43.56	90.22	25.40	94.21
Local-Prompt (w/o outlier)	8.63	98.07	23.23	95.12	31.74	92.42	34.50	92.29	**24.52**	**94.48**

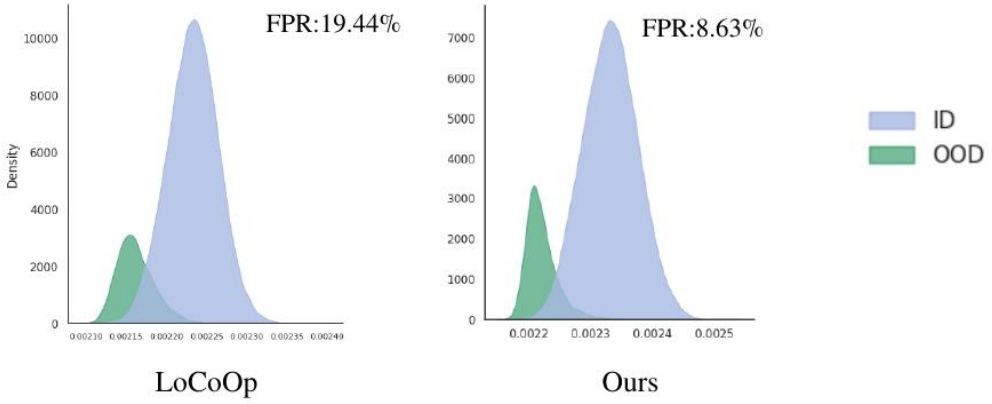

Figure 3: Comparison of density maps on iNaturalist. Local-Prompt localizes more discriminative regions that distinguish OOD species from ID classes, while global methods produce diffuse attention across the entire image.

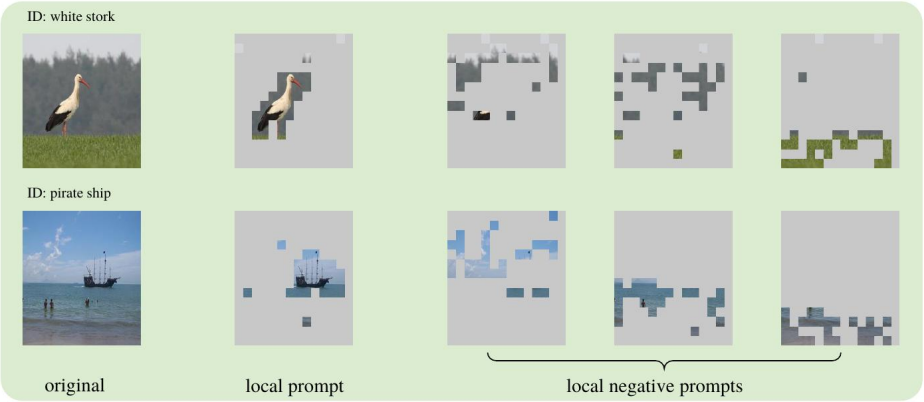

Figure 4: Visualization of ID and OOD related regions captured by the trained local prompts (green: ID-relevant; red: OOD-relevant). Local prompts successfully identify the discriminative regions that distinguish ID from OOD.

Ablation on OOD Score Strategies. Figure 6 ablates the choice of OOD score strategy. Using only global MCM, only local score, or naive combinations all perform worse than the Regional-MCM formulation with negative prompts included. The inclusion of local negative prompts in the normalization denominator of the regional score is critical for accurate OOD detection.

Ablation on Negative Augmentation Strategy. Table 7 compares different augmentation strategies for generating hard negatives. Without any augmentation (Baseline), FPR95 is 28.97. Cutout augmentation hurts performance (33.21 FPR95), while Gaussian noise also degrades results (30.76 FPR95). Random crop achieves the best result (28.77 FPR95) because it naturally simulates the appearance of partial/background regions that correspond to OOD-like local content. The semantic filtering via global prompts is key to identifying the most informative crops.

Ablation on Number of Local Negative Prompts. Table 8 examines the effect of N_{neg}. The performance is relatively stable across $N_{\text{neg}} \in \{100, 200, 300\}$, with $N_{\text{neg}} = 200$ giving the best average FPR95 of 28.52. The robustness to this hyperparameter suggests that the diversity regular-

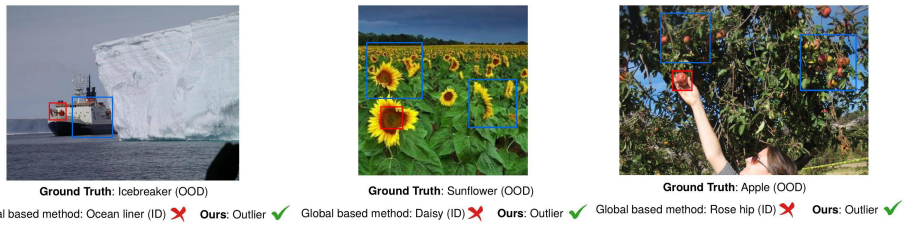

Figure 5: Examples of outliers from the SUN dataset that fail to be detected by previous global-based methods. Our method with local enhancement successfully discriminates them based on subtle differences in certain regions.

Table 6: Effectiveness of each component in loss function.

Loss	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
Baseline ($\mathcal{L}_{\text{pos}}$)	11.64	97.59	26.81	94.59	34.50	91.59	50.46	87.84	30.85	92.90
$+\mathcal{L}_{\text{neg}}$	9.84	97.88	24.37	94.97	32.84	91.94	50.02	88.32	29.27	93.27
$+\mathcal{L}_{\text{neg}} + \mathcal{L}_{\text{reg}}$	9.39	97.97	24.20	95.10	32.13	92.21	49.37	88.66	**28.77**	**93.48**

ization successfully prevents over-specialization for any specific N_{neg} value. We use $N_{\text{neg}} = 200$ in all other experiments.

6 Conclusion

We presented Local-Prompt, a coarse-to-fine few-shot OOD detection framework that decomposes prompt learning into frozen global prompts and trainable local prompts. By combining Global Prompt Guided Negative Augmentation with Local Prompt Enhanced Regional Regularization, Local-Prompt trains local prompts to capture the subtle regional cues that distinguish hard OOD samples from ID data—without requiring any external OOD data. The Regional-MCM score integrates global and local evidence through both local positive and negative prompts for robust inference. Local-Prompt is orthogonally extensible: plugging in trained global prompts from existing methods provides consistent further gains. Extensive experiments demonstrate state-of-the-art performance across large-scale standard and near-OOD benchmarks, with Local-Prompt achieving 4-shot results that surpass prior 16-shot baselines and competitive performance against methods that use powerful external outlier samples. We believe the coarse-to-fine paradigm established by Local-Prompt opens a productive direction for future work on efficient and robust OOD detection.

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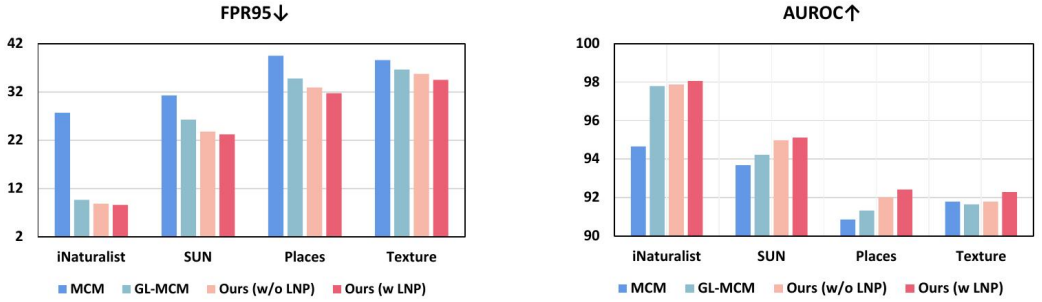

Figure 6: Ablation study of different OOD score strategies. Regional-MCM combining global and local scores achieves the best performance.

Table 7: Influence of different negative augmentation strategies.

	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
Baseline	11.18	97.68	23.30	95.25	30.97	92.44	50.44	87.89	28.97	93.31
+cutout	17.47	96.64	26.17	94.73	34.38	91.84	54.84	86.88	33.21	92.52
+gaussian noise	11.32	97.59	25.70	94.39	34.99	91.71	51.06	88.15	30.76	92.96
+random crop	9.39	97.97	24.20	95.10	32.13	92.21	49.37	88.66	**28.77**	**93.48**

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Table 8: Influence of different number of local negative prompts.

N_{neg}	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
100	9.02	98.00	24.71	94.82	32.77	91.87	48.81	88.50	28.82	93.30
200	9.65	97.90	22.58	95.26	32.03	92.18	49.85	88.00	28.52	93.33
300	9.39	97.97	24.20	95.10	32.13	92.21	49.37	88.66	28.77	93.48

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